

Second Quantisation for General Infinitely Divisible Measures: A Scoped Later Answer

Literature status for arXiv:1305.5146

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Status: literature already answered, with a scope correction. Applebaum–van Neerven’s follow-up arXiv:1405.1276 explicitly completes the Gaussian/pure-jump programme for general infinitely divisible measures when the skew factor is infinitely divisible. The same follow-up proves that this factor hypothesis is not automatic.

Source conjecture

For a skew map $T : E_1 \rightarrow E_2$ with

$$T(\mu_1) * \rho = \mu_2,$$

the source constructs a contraction

$$(P_T f)(x) = \int_{E_2} f(Tx + y) d\rho(y)$$

and identifies it with symmetric second quantisation when the endpoint laws are both Gaussian (Theorem 3.5) or both infinitely divisible of pure-jump type (Theorem 4.4) [1].

Remark 4.5 on source PDF page 14 asks whether these theorems extend when μ_1 and μ_2 are arbitrary infinitely divisible measures.

Exact later answer

The same authors’ separate follow-up explicitly cites the source and states on supporting PDF page 2:

In this article, we complete the programme by dealing with the case where the λ_i are general infinitely divisible measures.

It decomposes each law canonically as

$$\lambda_i = \gamma_i * \pi_i,$$

where γ_i is centred Gaussian and π_i is the drift-plus-pure-jump part. Proposition 2.1 shows that, if the skew factor is infinitely divisible, the skew relation splits into Gaussian and pure-jump skew relations. The two known second-quantisation maps can therefore be tensored.

Theorem 3.7 on supporting PDF page 10 gives the resulting commuting diagram. Its one-particle Hilbert space is

$$H_{\gamma_i} \oplus L^2(E_i, \nu_i),$$

and its Fock-space arrow is

$$\Gamma(T^*) = \bigoplus_{n=0}^{\infty} (T^*)^{\odot n}.$$

The transition operator P_T is intertwined with this arrow after embedding f as $F_f(x, y) = f(x + y)$ into the independent Gaussian–jump product space. This is precisely the mixed extension suggested in the source [2].

Why the scope qualifier matters

Theorem 3.7 uses the standing assumption that the skew factor ρ is infinitely divisible. Endpoint infinite divisibility alone does not imply this. Example 2.2 (Rosiński) on supporting PDF page 4 constructs an infinitely divisible law μ_1 , a non-infinitely-divisible law ρ , and an infinitely divisible law $\mu_2 = \mu_1 * \rho$.

Accordingly, the later paper gives an explicit affirmative answer to the natural infinitely-divisible-factor formulation and identifies a genuine obstruction to deleting that qualification. It does not assert that the unrestricted arbitrary-factor wording follows merely because μ_1 and μ_2 are infinitely divisible. This status note therefore records a scoped literature answer rather than overstating Theorem 3.7.

Verification and provenance

The source conjecture was checked on PDF page 14. The follow-up’s explicit programme statement was checked on PDF page 2, Proposition 2.1 and the Rosiński example on pages 3–4, and the final mixed theorem on PDF page 10. The follow-up is by the same authors, cites the source as its previous paper, and explicitly describes its result as completing that programme. The relation is therefore author-explicit, not an agent-only theorem match.

References

- [1] D. Applebaum and J. van Neerven, *Second quantisation for skew convolution products of measures in Banach spaces*, Electron. J. Probab. **19** (2014), paper 11, 1–17; arXiv:1305.5146.
- [2] D. Applebaum and J. van Neerven, *Second quantisation for skew convolution products of infinitely divisible measures*, Infin. Dimens. Anal. Quantum Probab. Relat. Top. **18** (2015), 1550003; arXiv:1405.1276.